

Puen Publishing & Regency Press — Executive Project Handover Document

Project: Puen Publishing E-Commerce & Gamification Platform

Live Production URL: <https://puenpublishing.com>

Client: Ray Puen (Regency Press / Puen Publishing / New Life Mission Board, Inc.)

Lead Engineer: Huzaifah

Version / Milestone: Phase 1 Handover (October 2026)

Section 1: Executive Summary & What Was Built

The platform at **puenpublishing.com** serves as the digital storefront, institutional distribution hub, and gamified reading adventure for Ray Puen's children's literature venture: *The Geezy Goober's Guide to Icky Island (The Search for the Magic Pen, Volume 1)*.

Key capabilities delivered:

- **Direct-to-Consumer Storefront:** High-converting book showcase featuring the authentic, high-resolution Book 1 front cover art (Nepal Recovery Initiative Edition with gold seal), real-time pricing, and seamless checkout.
 - **19-Slot Digital Gamification Engine (/dashboard/book2):** Real-time interactive virtue badge album (Humility, Joy, Perseverance, Integrity, Courage, etc.) backed by Supabase cloud storage. Tracks quiz scoring (400 points) and peer referrals (300 points) to unlock Volume 2 rewards.
 - **Stakeholder Demo Override (/dashboard/book2?demo=1):** A dedicated, unrestricted presentation mode for publisher meetings that instantly showcases the fully illuminated sticker ledger and completion state without prerequisite quizzes.
 - **Chief Scout Patrol Leader Funnel (/guide):** A \$10 digital bundle generating unique Patrol Hub tokens and three gift invitation codes, with automated buyer email delivery and an administrative one-click approval gate for printed rewards.
 - **Institutional Wholesale Portal (/institutions):** Flat-rate wholesale sponsorship tiers (\$40 / \$200 / \$400) tailored for preschools, church ministries, and homeschool networks with 1:1 student matching.
 - **Video Portal Destinations (/v1 through /v5):** Permanently printed QR/URL destinations embedded in physical books. Displays embedded video players when connected, or a child-friendly status message when pending.
 - **Production Email & Webhook Infrastructure:** Fully authenticated custom transactional email engine via Resend (send.puenpublishing.com) and real-time failure alerting via Google Apps Script relay.
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Section 2: Architecture & Technology Stack

The platform is engineered as a modern, decoupled cloud application with high resilience and serverless execution:

Layer	Technology	Purpose & Implementation Details
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Frontend Framework	Next.js 14 (App Router)	Server Components + Client hydration, React 18, Tailwind CSS.
Hosting Platform	Vercel Serverless Edge	Global CDN distribution, automatic HTTPS, branch previews, zero-downtime deployments.
Database	Supabase (PostgreSQL)	Persistent storage for scout profiles, progress arrays, patrol leader tokens, and orders.
Payment Gateway	Stripe API	Secure hosted checkout sessions for retail, patrol bundles, and wholesale packages.
Transactional Email	Resend API (HTTP)	One-to-one transactional notifications with strict timeout guards and secret scrubbing.
DNS & Mail Routing	Namecheap DNS	SPF, DKIM, and MX records configured on dedicated subdomain <code>send.puenpublishing.com</code> .
Incident Alerting	Google Apps Script	Webhook relay dispatching revenue-critical server alerts directly to administration Gmail.
Automated Testing	Vitest (Node Environment)	325 strict automated tests verifying math, auth boundaries, timeouts, and redaction.

Section 3: Credentials & Access Architecture

All sensitive credentials and API tokens are managed strictly through environment variables. Zero credentials are committed to the public Git repository.

Environment Variable Map

Variable Key	Scope	Description & Location
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NEXT_PUBLIC_SITE_URL	Client / Server	Canonical public URL (https://puenpublishing.com).
NEXT_PUBLIC_SUPABASE_URL	Client / Server	Public API endpoint of the Supabase PostgreSQL project.
NEXT_PUBLIC_SUPABASE_ANON_KEY	Client / Server	Public anonymous API key for client-side queries.
SUPABASE_SERVICE_ROLE_KEY	Server Only	High-privilege key used for secure server-side mutations.
STRIPE_SECRET_KEY	Server Only	Secret API key for creating Stripe checkout sessions.
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY	Client / Server	Stripe publishable key for client checkout redirect.
STRIPE_WEBHOOK_SECRET	Server Only	Signing secret used to verify webhook event integrity.
RESEND_API_KEY	Server Only	Bearer token for authenticating with the Resend email API.
EMAIL_FROM	Server Only	Sender address: The Chief Explorer .
ADMIN_EMAIL	Server Only	Recipient for free guide claims (info@puenpublishing.com).
ADMIN_TOKEN	Server Only	Secret string (16+ chars) protecting /admin and detailed /api/health .
ALERT_WEBHOOK_URL	Server Only	Google Apps Script endpoint receiving emergency incident logs.

Section 4: Production Deployment & Server Runbook

Deployment Procedure

The project is connected to Vercel via GitHub continuous integration. Every push to the `main` branch automatically builds and deploys to production:

- **Local Checks Before Pushing:**

```
# 1. Typecheck
npx tsc --noEmit

# 2. Code Linting
npm run lint

# 3. Automated Test Suite
npx vitest run
```

- **Deploying via Git:**

```
git add <modified_files_by_name>
git commit -m "feat/fix: descriptive message"
git push origin main
```

- **Vercel Manual Redeploy:**

- Navigate to vercel.com -> `raypuenpublishing` project.
- Go to **Deployments** tab.
- Click the three dots `...` on the active deployment and select **Redeploy**.

- **Domain Routing & Canonical 308 Redirect:**

- Primary Domain: `puenpublishing.com`
- Secondary Domain: `www.puenpublishing.com` -> redirects to `puenpublishing.com`
- Platform Domain: `raypuenpublishing.vercel.app` -> 308 permanent redirect to `puenpublishing.com`.

Section 5: Database Schema & Migration Status

The database is structured into four core relational tables within Supabase PostgreSQL:

- **scout_profiles :**
- `token` (Text, Primary Key): Unique scout identifier (e.g., `SC-xxxx` or `CAPTAIN-RAY-700`).
- `scout_name` (Text): Child's name.
- `completed_pages` (Integer Array): Array of registered pages (\$1..19\$).
- `quiz_score` (Integer): Score achieved on comprehension quiz (\$0..400\$).
- `referral_score` (Integer): Score achieved from recruited friends (\$0..300\$).
- `status` (Text): Progression tier (`In_Progress` , `Academic_Pass` , `Unlock_Volume_2`).
- `friends_completed` (Integer): Counter of friends who registered page 1 (\$0..3\$).
- `book3_sponsored` (Boolean): Flag unlocked when relative sponsors Book 3 (\$40).
- **patrol_leaders :**
- `id` (UUID, Primary Key): Unique patrol identifier.
- `email` (Text): Buyer's email address.
- `token` (Text): Secret access token for Patrol Hub.

- `gift_codes` (Text Array): Array of three unique gift codes issued to buyer.
- `registered_families` (Integer): Number of redeemed gift codes.
- `claimed_printed_guide` (Boolean): Claim status for physical guide reward.
- **patrol_gifts :**
- `code` (Text, Primary Key): Unique gift voucher code (`GP-xxxx`).
- `patrol_id` (UUID): Reference to parent patrol leader.
- `redeemed_by` (Text): Email of recipient family who claimed the code.
- `redeemed_at` (Timestamp): Exact time of activation.
- **videos :**
- `slug` (Text, Primary Key): URL identifier (`v1` , `v2` , `v3` , `v4` , `v5`).
- `title` (Text): Display title of video episode.
- `youtube_id` (Text): YouTube Unlisted ID (when empty, displays polite waiting card).
- `status` (Text): `live` or `pending` .

Migration Status: All migrations (`20260926_add_scout_referrals.sql` , `20260926_create_patrol_and_guides.sql` , and `videos` table) are fully applied and verified live via `/api/health` .

Section 6: Background Tasks & Automation Architecture

The platform runs purely on serverless event-driven architecture, avoiding fragile server cron daemons:

- **Stripe Webhook Event Processing (`/api/webhooks/stripe`):**
- Listens for `checkout.session.completed` events.
- Automatically discriminates order type (`retail` , `patrol_bundle` , `sponsor_volume_3` , `wholesale`).
- Automatically generates gift codes and emails buyer receipt with zero human intervention.
- **Transactional Email Dispatch:**
- Sent through Resend HTTP endpoint with strict 8-second execution deadlines to prevent serverless execution stalls.
- Automatic fallback alerting to admin email if webhook transport fails.
- **Crash & Error Alerting:**
- Revenue-critical paths route unexpected failures to `alertFailure()` .
- Dispatches incident notifications to Google Apps Script relay, landing directly in the administrative inbox.

Section 7: Verification & Test Suite Results

The codebase contains a comprehensive automated test suite enforced through Vitest:

- **Total Test Files:** 37 test files passing (100% clean).
- **Total Automated Tests:** 325 tests passing.
- **Typecheck Status:** `npm tsc --noEmit` exits with **0 errors**.
- **Linting Status:** `npm run lint` exits with **0 errors**.

Key Test Coverage Highlights:

- **Gamification Arithmetic (`lib/gamification.test.ts`):** Verifies 400-point quiz scoring, 300-point friend logic, and the 700-point Volume 2 unlock boundary.
 - **Authorization & Security (`lib/scout-access.test.ts`):** Proves scout token ownership, constant-time token comparison, and cookie security.
 - **Patrol Funnel (`lib/patrol.test.ts`):** Verifies receipt generation, 3-gift-code issuance, and approval gate security.
 - **Redaction Hygiene (`lib/redact.test.ts`):** Proves customer shipping addresses and authentication tokens are scrubbed from server logs.
 - **API Endpoints (`app/api/ .test.ts`):*** 100% route coverage testing successful orders, network timeouts, and invalid payloads.
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Section 8: Support & Maintenance Guidelines

Quick Troubleshooting Cheat Sheet

- **A Customer Says They Didn't Receive Their \$10 Patrol Receipt Email:**
 - Log in to resend.com/emails -> Check the delivery log for the customer's email.
 - The link can also be looked up directly from their Stripe session ID via the database.
 - **Adding a New Video to `/v1` through `/v5` :**
 - No code deployment is needed.
 - Open Supabase Dashboard -> Table Editor -> `videos` table.
 - Update the row for `v1` with the YouTube Unlisted video ID and set `status` to `live` .
 - **Approving a Family's Free Printed Parent's Guide:**
 - Visit `https://puenpublishing.com/admin` .
 - Enter your `ADMIN_TOKEN` when prompted.
 - Review pending claims and click **Approve** to record fulfillment.
 - **Switching Stripe from Test Mode to Live Mode:**
 - Go to Vercel -> Settings -> Environment Variables.
 - Replace `pk_test_...` and `sk_test_...` with your `pk_live_...` and `sk_live_...` keys from the Stripe Dashboard.
 - Add the live `STRIPE_WEBHOOK_SECRET` (`whsec_...`).
 - Trigger a Redeploy in Vercel.
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Handover document compiled and verified for Regency Press & Puen Publishing — October 2026.